RESEARCH MANUSCRIPT

Pose-Assisted Exact-Capacity Transductive Recognition of Unseen sEMG Hand Gestures from Basic-Movement Calibration

A fully symmetric subject-level evaluation from NinaPro DB5 basic finger and hand open-close calibration to unseen Exercise-B gestures

Final pre-submission manuscript for author verification

2026-08-15

NinaPro DB5 pose-assisted unseen-gesture manuscript

Title Page

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Running title: Pose-assisted transductive unseen-gesture recognition

Article type: Original research

Dataset: NinaPro Database 5 (DB5) only

Abstract

Background

Surface electromyography (sEMG) supports non-invasive wearable hand-gesture interfaces, but inter-person signal variation and the need to record every gesture from each user limit generalization. Hand pose can provide an auxiliary semantic space linking muscular activity to gesture structure, provided that the use of target labels, target kinematics, and query-set information is reported explicitly.

Objective

We evaluated the internal performance of a Target-A stability-selected pipeline that used a target person's 12 basic Exercise-A movements plus Exercise-B hand open and close (B5/B6) to recognize B1/B2/B3/B4/B7 without labeled target examples from those evaluated gestures.

Methods

Each of the ten NinaPro DB5 participants served once as the target. In every fold, pose recordings from the other nine participants provided B1/B2/B3/B4/B7 semantic prototypes. Target A1-A12+B5/B6 EMG and glove data were permitted for calibration, whereas evaluated-query inference received target EMG only; target query glove and labels remained inaccessible until prediction was complete. Non-overlapping 2-s windows yielded 128 time-domain EMG features and 66 pose descriptors. A weighted RidgeCV bridge predicted pose from EMG. Twenty-seven candidates, formed by three EMG views, three predicted-pose weights, and three robust aggregation rules, were selected per target using 12 whole-class target-A holdouts and a B5/B6 eligibility criterion. The final query was partitioned into five clusters of exactly six trials and named through other-nine pose prototypes and Hungarian one-to-one assignment. Trial balanced accuracy (BA) was the principal metric.

Results

The final pipeline achieved mean trial BA and macro F1 of 0.8733 and window BA of 0.8730 across s1-s10. Subject-level trial BA ranged from 0.6000 to 1.0000, with perfect classification for six targets. Recall for B1/B2/B3/B4/B7 was 1.0000/0.7833/0.7833/0.9167/0.8833.

Conclusions

Unseen Exercise-B gestures were separable in a closed-catalog, pose-assisted transductive setting using target basic-movement and B5/B6 calibration. The final pipeline was reproduced through a locked execution and independent metric replay. Because earlier analyses of the same DB5 cohort informed candidate-family development, 0.8733 is internal development performance rather than an independent validation estimate. The findings establish a bounded full-batch result, not online inductive zero-shot recognition.

Keywords: surface electromyography; hand gesture recognition; NinaPro DB5; transductive learning; zero-shot learning; pose semantics; balanced clustering; subject-specific stability selection

1. Introduction

Surface electromyography records electrical manifestations of muscle activation non-invasively and has been extensively studied for control of upper-limb prostheses [1,2]. Laboratory gesture classifiers can perform well under controlled conditions, yet electrode placement, electrode-skin contact, contraction intensity, limb posture, recording session, and user physiology alter the observed signal distribution [1,2]. These factors make direct transfer across people difficult and sustain the practical burden of collecting labeled repetitions of every intended gesture from every user.

One route to lower that burden is to calibrate a limited vocabulary of elementary movements and infer other hand configurations without labeled target examples from the evaluated classes. EMG alone, however, does not explicitly encode how finger and joint configurations relate semantically. Synchronized hand pose provides a structured complementary modality that can organize gesture relationships. EMBridge recently pursued this principle with separately pretrained EMG and pose masked autoencoders, a Q-Former, instance contrast, masked pose reconstruction, and community-aware soft contrastive learning [5]. That work motivates pose-guided unseen-gesture recognition but does not imply that a large cross-modal neural architecture will be optimal at every dataset scale or under every evaluation protocol.

The present study addresses the same broad scientific question with an explicit structured pipeline. NinaPro DB5 provides synchronized 16-channel EMG from two Myo armbands and 22-sensor CyberGlove recordings from ten intact participants, each performing six repetitions of 52 movements [3,4]. We used the 12 basic finger movements in Exercise A and Exercise-B hand open and close (B5/B6) for target calibration, and retained B1/B2/B3/B4/B7 as evaluated gestures. This is an

operational semantic split asking whether primitives and two whole-hand configurations support discrimination of other multi-finger configurations. It is not a universal physiological claim that every Exercise-B gesture is intrinsically more complex than every Exercise-A movement.

The study has two linked questions. First, can target EMG from evaluated gestures be clustered and named using other-person pose semantics when no labeled target EMG examples from those gestures are available? Second, if no single EMG representation is equally stable for all users, can permitted target primitives select a stable representation without reading evaluated-query outcomes? We combined a weighted EMG-to-pose Ridge bridge, exact-capacity balanced clustering, other-nine pose prototypes, one-to-one semantic naming, and a target-A stability selector over a fixed 27-candidate family.

The main contributions are fourfold. First, all s1-s10 participants were included as equal target folds, with the current target excluded from the pose-donor pool in every fold. Second, the information boundary distinguishes target calibration, other-person pose semantics, unlabeled target-query EMG, and evaluation-only outcomes. Third, one candidate family and one stability rule were applied to every target. Fourth, cohort metrics are accompanied by all subject-level results, class recall, subject-by-class error structure, and an explicit statement that the estimate is internal development performance.

2. Related Work

2.1 sEMG pattern recognition and user variability

Classical myoelectric pattern recognition summarizes short windows with time-domain descriptors such as mean absolute value, waveform length, zero crossings, and slope-sign changes [7]. Redundancy among candidate EMG descriptors has motivated systematic feature reduction and selection [8]. Together with known user and acquisition variability [1,2], this supports evaluating prespecified feature families rather than assuming that one fixed subset is optimal for every person.

The DB5 acquisition study demonstrated recognizable movement structure with the double-Myo setup and reported significant differences associated with movements and participants [3]. We therefore retained the participant, rather than the trial or window, as the unit of generalization and statistical interpretation.

2.2 Pose-guided cross-modal learning

EMBridge first pretrains EMG and pose encoders with masked autoencoding, then uses a Q-Former to extract pose-informative queries while retaining a frozen pose anchor [5]. Its instance-level contrast aligns paired EMG and pose; masked pose reconstruction requires EMG-derived queries to recover missing pose content; and community-aware contrast preserves neighborhood geometry from pose space. Our work shares the use of pose as semantic guidance but does not implement the EMBridge neural architecture or its losses. Instead, time-domain EMG features, a linear pose

bridge, balanced clustering, and prototype naming expose where each information source enters the prediction.

2.3 Transductive recognition of unseen classes

Zero-shot learning uses auxiliary semantic structure to recognize target classes without labeled target examples. Transductive zero-shot methods may additionally exploit the structure of unlabeled target observations rather than classifying each observation independently [6]. Our method jointly clusters all 30 target trials and therefore is transductive. It also knows that the batch contains five catalog classes with exactly six trials per class. The resulting task is stronger in side information than online or open-catalog zero-shot classification.

2.4 Capacity-constrained clustering and semantic assignment

Balanced k-means can enforce specified cluster sizes by solving an assignment problem during clustering [10]. We use the known six-repetition DB5 protocol as an exact cluster-capacity constraint. Anonymous clusters are subsequently linked one-to-one to pose class prototypes. Separating partitioning from semantic naming allows partial diagnosis of whether errors arise from EMG-side separation or from cluster-to-class correspondence.

3. Materials and Methods

3.1 Study design

This study was a secondary computational analysis of the publicly released, de-identified NinaPro DB5 dataset. All ten participants, s1-s10, were included. Each participant served once as target in a fully symmetric ten-fold evaluation. In every fold, the current target was excluded from its pose-donor pool and the other nine participants served as donors. The same rule was used for every target.

The objective was to characterize internal B1/B2/B3/B4/B7 recognition performance for the final pipeline combining target A1-A12+B5/B6 calibration, other-nine pose semantics, and target-A stability selection. This was a retrospective development study in the same DB5 cohort, not an independent external validation study.

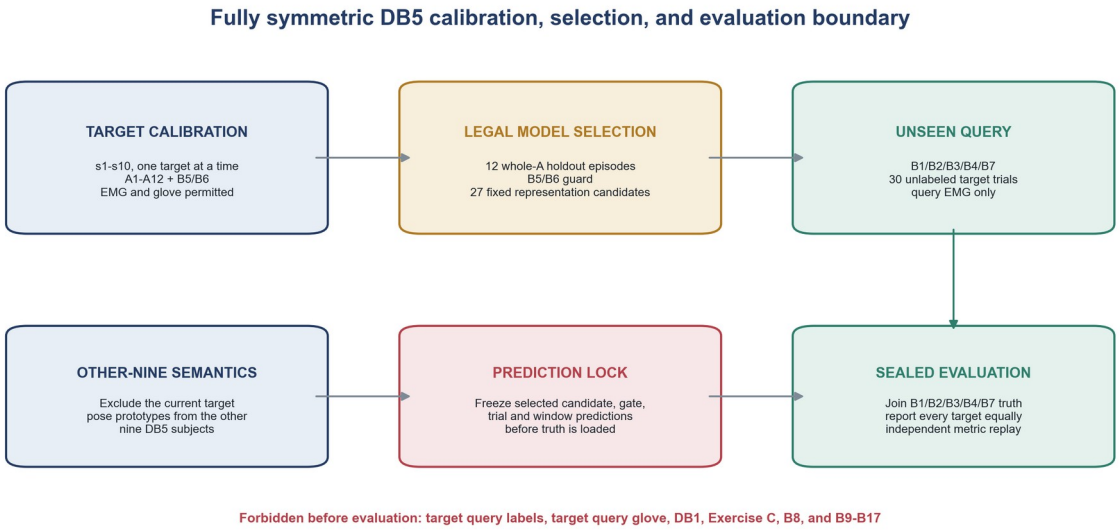


Figure 1. Fully symmetric subject-level design and information boundary. The current target is excluded from the pose-donor set. Candidate selection uses target A1-A12+B5/B6 and other-nine pose semantics. Target B1/B2/B3/B4/B7 glove and labels are unavailable until predictions are sealed.

3.2 Dataset and movement roles

DB5 contains ten intact participants, 16 channels of EMG sampled at 200 Hz from two Myo armbands, 22-sensor CyberGlove recordings, 52 movements, and six repetitions per movement [3,4]. Corrected movement and corrected repetition labels were used.

Under the NinaPro movement definitions [3,4,12], Exercise A comprises A1 index flexion, A2 index extension, A3 middle-finger flexion, A4 middle-finger extension, A5 ring-finger flexion, A6 ring-finger extension, A7 little-finger flexion, A8 little-finger extension, A9 thumb adduction, A10 thumb abduction, A11 thumb flexion, and A12 thumb extension. The Exercise-B classes used here are B1 thumb up; B2 extension of the index and middle fingers with flexion of the others; B3 flexion of the ring and little fingers with extension of the others; B4 thumb opposition to the base of the little finger; B5 abduction of all fingers; B6 fingers flexed together in a fist; and B7 pointing with the index finger. B5/B6 were target calibration classes, whereas B1-B4/B7 were the unseen evaluation classes.

Table 1. Data roles and information boundary

Data	Stage	Role
Target A1-A12 EMG and glove	Calibration and selection	Preprocessing, Ridge bridge, whole-class A holdouts
Target B5/B6 EMG and glove	Calibration and selection	Hand open-close calibration and alternative-candidate eligibility

Data	Stage	Role
Other-nine A1-A12, B5/B6, and B1/B2/B3/B4/B7 glove	Semantic reference	Episode-specific held-class pose prototypes
Target B1/B2/B3/B4/B7 EMG	Evaluation input	Unlabeled 30-trial query batch
Target evaluated glove and labels	Outcome computation only	Joined after prediction sealing
Five-class catalog and six trials/class	Evaluation prior	Exact-capacity clustering and one-to-one naming
DB1, Exercise C, B8, and B9-B17	Excluded	No model-selection or manuscript evidence

The target provides zero labeled EMG examples from B1/B2/B3/B4/B7. Nevertheless, the method uses target B5/B6 EMG and glove, other-nine evaluated-class glove, the complete query batch, and exact class frequencies. We therefore describe the task as **target-calibrated, pose-assisted transductive recognition of unseen gestures**.

3.3 Signal processing and windows

Non-rest corrected-movement segments were separated by corrected repetition. Non-overlapping windows of 2 s (400 samples) were extracted at 200 Hz. Every complete window contained 16 EMG channels and 22 glove channels. EMG was processed with a fourth-order 2-95 Hz Butterworth band-pass filter and a 60 Hz notch filter (Q=30). Fit-dependent transforms were estimated from permitted calibration data only.

Eight descriptors were computed for each EMG channel: mean absolute value (MAV), root mean square, standard deviation, peak-to-peak amplitude, raw waveform length, zero-crossing rate, slope-sign-change rate, and normalized waveform length. The resulting 128-dimensional vector extends established low-compute time-domain EMG feature families [7,8].

For each glove sensor, we concatenated the window mean, standard deviation, and the difference between the final and initial 20% means, yielding a 66-dimensional pose descriptor. The DB5 glove channels were not treated as calibrated absolute anatomical joint angles [4]; they were used only as relative sensor-space pose descriptors.

3.4 Weighted Ridge EMG-to-pose bridge

An L_2-regularized multi-output Ridge model mapped the 128-dimensional EMG vector to the 66-dimensional pose descriptor [9]. Calibration windows were weighted so that every participant × exercise × class × repetition group contributed equal total weight. Separate StandardScaler transforms were fitted for EMG and pose. RidgeCV selected α from 13 logarithmically spaced values from 10^{-2} to 10^{4}.

$$J(B) = \Sigma_i \, w_i \, \|y_i - B^T x_i\|_2^2 + \alpha \, \|B\|_F^2$$

$$\hat{B} = \arg\min J(B)$$

Here, x_i is an EMG feature vector, y_i is the pose descriptor, and w_i is the group-balancing weight. At evaluated-query inference, predicted pose $\hat{y}_i$ was computed from EMG; actual target query glove was not accessed.

3.5 Trial representations and exact-capacity clustering

Window features belonging to one trial were averaged and L2-normalized. Three EMG views were available:

- **MAV16:** one MAV value per channel;
- **amplitude64:** MAV, root mean square, standard deviation, and peak-to-peak amplitude per channel;
- **all128:** all eight descriptors from all channels.

An EMG trial vector e_t and predicted-pose vector q_t were concatenated with pose weight λ:

$$c_t = [e_t \; ; \sqrt{\lambda}\, q_t] \,/\, \sqrt{(1 + \lambda)}$$

The 30 evaluated trials for a target were partitioned into five clusters of exactly six trials. Six slots were created for each centroid. Linear assignment under squared Euclidean cost alternated with centroid updates until convergence, following the assignment-based principle of balanced k-means [10]. The lowest within-cluster sum of squares over 64 deterministic k-means++ initializations was retained.

$$J(z, \mu) = \Sigma_t \, \|c_t - \mu(z_t)\|_2^2$$

$$(\hat{z}, \hat{\mu}) = \arg\min J(z, \mu)$$

$$\text{subject to } \Sigma_t \, I(z_t = j) = 6, \;\; j \in \{1, \ldots, 5\}.$$

The exact-capacity constraint uses the DB5 acquisition protocol and is a material source of side information. It does not generalize automatically to batches with unknown or unequal class frequencies.

3.6 Other-nine pose prototypes and cluster naming

For each target, 66-dimensional class prototypes for B1/B2/B3/B4/B7 were formed by averaging glove descriptors from the other nine participants. Cosine similarities between EMG-predicted pose and the five prototypes were aggregated within each anonymous cluster, creating a 5 × 5 cluster-class score matrix. Hungarian one-to-one assignment mapped clusters to distinct class names [10].

The fixed naming rule compared one-to-one assignments from all 66 pose dimensions with assignments from 44 dynamic dimensions that excluded the 22 window means. For either score matrix, the assignment margin was the total score of the best cluster-class permutation minus that of

the second-best permutation. A gate fixed before evaluated-query prediction selected raw naming for s3-s6, s8, and s10. For s1, s2, s7, and s9, dynamic naming was used only when the raw and dynamic mappings differed, the dynamic margin was larger, and the raw margin was below 0.08. Target-A selection reused these gates without retuning.

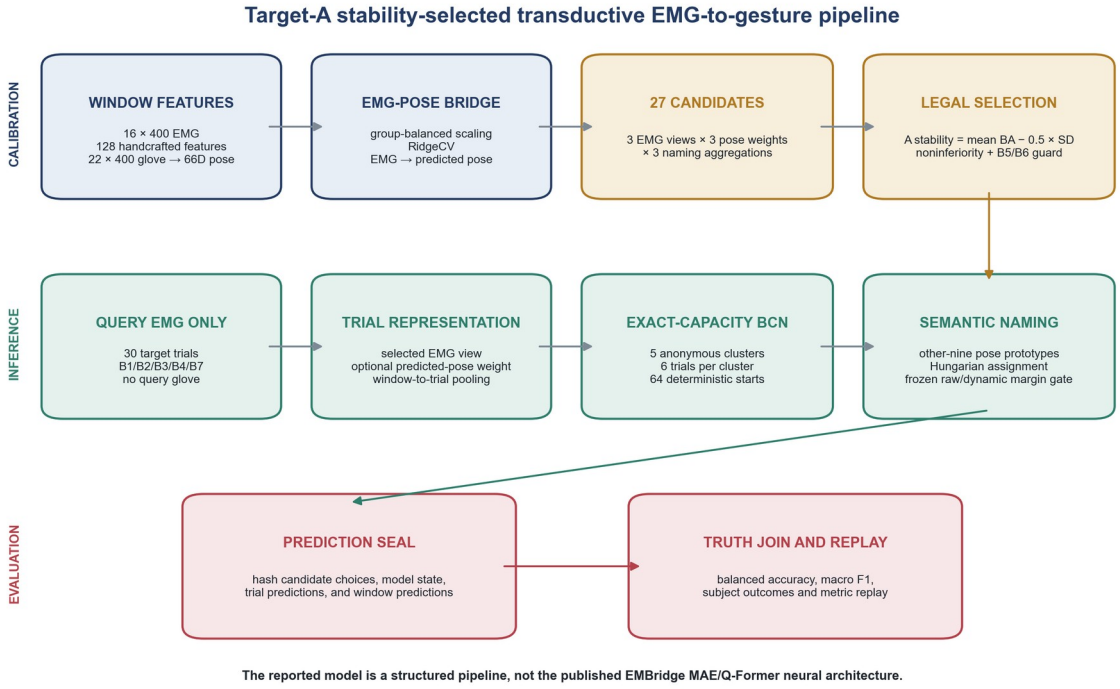


Figure 2. Common subject-specific pipeline. The candidate family, stability formula, eligibility criteria, and evaluated-query procedure were identical for all s1-s10 folds. Selected configurations could differ with permitted target calibration data.

3.7 Target-A stability selection

The selector used the complete Cartesian product of:

1. EMG clustering view: MAV16, amplitude64, or all128;
2. predicted-pose clustering weight: 0, 0.25, or 1;
3. within-cluster pose-score aggregation: mean, median, or one-per-tail trimmed mean.

The fixed family therefore contained 27 candidates.

Exercise-A classes were paired by finger or movement axis and opposing direction: A1/A2 (index flexion/extension), A3/A4 (middle-finger flexion/extension), A5/A6 (ring-finger flexion/extension), A7/A8 (little-finger flexion/extension), A9/A10 (thumb adduction/abduction), and A11/A12 (thumb flexion/extension). One pair was omitted per episode; one member was alternately taken from each

of the remaining five pairs to form a five-class query. Six omitted pairs and two phases produced 12 whole-class holdout episodes.

For candidate m, target-A stability was:

$$\text{Stability}(m) = \text{mean}_k[BA(m,k)] - 0.5 \times SD_k[BA(m,k)],\ \ k = 1, \ldots, 12.$$

An alternative was eligible only if all three conditions held:

1. mean target-A trial BA was no more than 0.005 below its prespecified default candidate;
2. target B5/B6 trial BA was at least 0.80;
3. stability exceeded the default candidate by at least 0.02.

Eligible candidates were ranked by stability, mean A trial BA, mean A window BA, B5/B6 BA, and fixed lexical candidate ID. If no alternative was eligible, the default candidate was retained even if its own B5/B6 BA was below 0.80; thus, the B5/B6 threshold constrained alternative selection rather than guaranteeing final B5/B6 performance. The rule was identical across targets and did not use B1/B2/B3/B4/B7 outcomes.

Table 2. Episode-specific fit, query, and semantic-prototype construction

Episode	Target fit for RidgeCV and scalers	Unlabeled target query	Other-nine pose prototypes	Candidate use
One of 12 A holdouts	The seven A classes not in that episode, plus B5/B6	Five held A classes, six trials/class	The same five held A classes; each donor was scaled using its legal episode fit	Stability and A non-inferiority
B5/B6 episode	All A1-A12; no target B5/B6 in fit	B5 and B6, six trials/class	Donor B5/B6; each donor was scaled from its A data	Alternative eligibility only
Evaluated B episode	Target A1-A12+B5/B6	B1/B2/B3/B4/B7, six trials/class	Donor B1/B2/B3/B4/B7	Locked prediction only

All fit-dependent RidgeCV and StandardScaler operations were re-estimated within each proxy episode. A class assigned to an episode query was absent from that episode's target fit. The final evaluated episode was run only after the subject-specific selection had been locked.

3.8 Information-access control and reproducibility

Candidate selection could read only permitted target A1-A12+B5/B6 data, other-nine pose semantics, and the fixed naming gate. Subject selections and the protocol were SHA-256 locked. A separate prediction stage loaded the locked selections and target evaluated-query EMG, generated trial and window predictions, and hash-locked them before outcome labels became accessible.

An independent reimplementation checked the following:

- all data paths were restricted to DB5;

- selections for all ten targets were reproducible;
- protocol, selection, and prediction hashes matched their lock records;
- prediction code had no access to the truth seal or evaluator;
- trial and window predictions matched the locked files; and
- every subject BA and the cohort means recomputed from raw predictions.

All checks passed.

3.9 Outcomes and statistical analysis

The principal outcome was trial balanced accuracy, defined as mean class recall [11]. Each target had five equally frequent classes, so trial accuracy and BA were numerically equal; BA was retained for class-oriented interpretation. Trial macro F1 and window BA were secondary. The evaluated set contained 300 trials and 388 non-overlapping windows. Window predictions were obtained by propagating each trial decision to its constituent windows; window BA is therefore a duration-weighted summary of trial decisions, not the output of an independent window classifier. Metrics were calculated per subject and then averaged with equal weight across s1-s10.

Performance was summarized using per-subject trial BA and macro F1, their cohort means, the subject range, and class recall. Because target folds shared overlapping nine-person donor banks and the same DB5 cohort informed method development, no confirmatory population confidence interval or hypothesis test for independent new users was reported.

Analyses used Python 3.12.8, NumPy 2.1.3, pandas 2.2.3, SciPy 1.15.1, and scikit-learn 1.8.0.

4. Results

4.1 Final cohort and subject-level performance

The final Target-A stability-selected pipeline achieved mean trial BA and macro F1 of 0.8733 and window BA of 0.8730 across s1-s10. With equal class frequencies, 262 of 300 trials were classified correctly. Subject-level trial BA ranged from 0.6000 to 1.0000. Targets s2, s3, s5, s7, s8, and s10 had all 30 trials classified correctly.

Table 3. Final subject-level trial balanced accuracy

Target	Final trial BA
s1	0.6000
s2	1.0000
s3	1.0000
s4	0.8667

Target	Final trial BA
s5	1.0000
s6	0.6000
s7	1.0000
s8	1.0000
s9	0.6667
s10	1.0000
Mean	**0.8733**

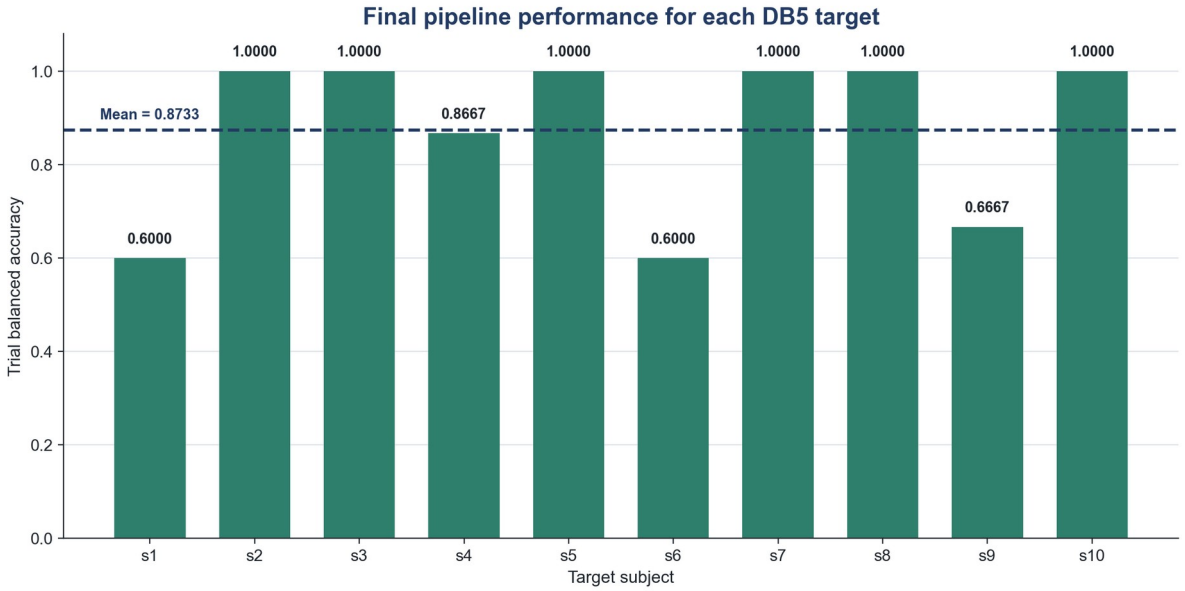


Figure 3. Trial balanced accuracy of the final pipeline for each target. The dashed line is the equal-weight s1-s10 mean of 0.8733. Every participant was included once as an equal target fold.

4.2 Target-A-selected subject configurations

Final configurations were selected using permitted target A1-A12+B5/B6 data only. Amplitude64 was selected for five targets, all128 for four, and MAV16 for one. Predicted pose was omitted from the clustering representation for five targets, used at weight 0.25 for three, and at weight 1 for two. Naming aggregation used the mean for seven targets, the median for two, and the trimmed mean for one.

Table 4. Final selected configuration and permitted selection metrics

Target	EMG view	Pose weight	Naming aggregation	A stability	Mean A trial BA	B5/B6 BA
s1	amplitude64	0	mean	0.3516	0.4361	1.0000

Target	EMG view	Pose weight	Naming aggregation	A stability	Mean A trial BA	B5/B6 BA
s2	MAV16	0	mean	0.4928	0.6250	0.0000
s3	all128	1	mean	0.5852	0.7306	1.0000
s4	all128	0.25	median	0.2961	0.4028	1.0000
s5	all128	0.25	median	0.4625	0.5667	1.0000
s6	amplitude64	1	mean	0.4579	0.5444	1.0000
s7	all128	0.25	trimmed mean	0.5142	0.6333	1.0000
s8	amplitude64	0	mean	0.7311	0.8389	1.0000
s9	amplitude64	0	mean	0.3866	0.5667	1.0000
s10	amplitude64	0	mean	0.7709	0.8694	0.0000

The same all128 + pose 0.25 + median configuration was independently selected for two targets, illustrating that final configurations could recur within the common candidate family. B5/B6 BA was 0.0000 for s2 and s10 because the 0.80 condition applied only to eligible alternatives and did not certify a retained default candidate.

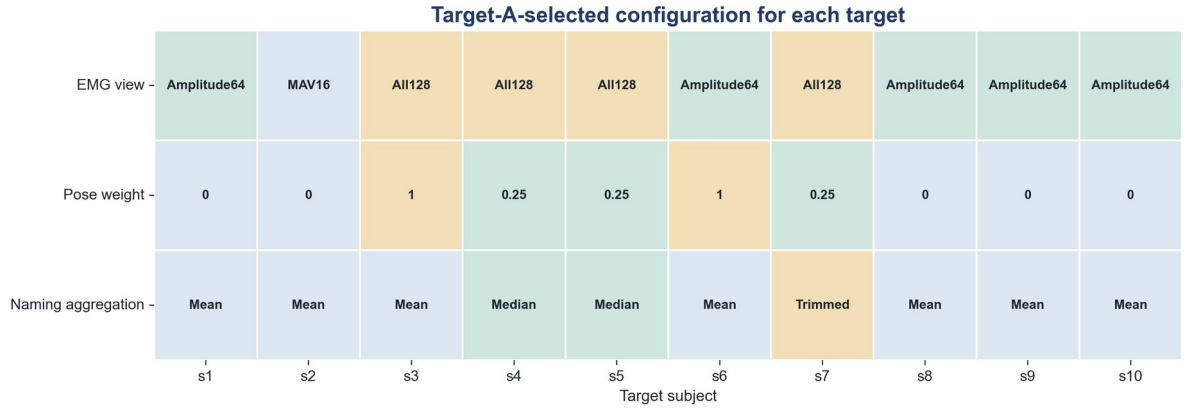


Figure 4. Final subject configuration selected by Target-A stability. The same 27 candidates and selection rule were applied to every target, while permitted calibration data could select different EMG views, predicted-pose weights, and naming aggregations.

4.3 Class recall and between-target error structure

Recall across all 300 trials was 1.0000/0.7833/0.7833/0.9167/0.8833 for B1/B2/B3/B4/B7. Each class contributed 60 trials. All 60 B1 trials were correct. Major residual errors involved reciprocal B2-B3 confusion, B3-B7 confusion, and B2-B4 confusion.

Errors remained for four targets: s1, s4, s6, and s9. S1 showed reciprocal B3-B7 confusion, s4 and s9 showed reciprocal B2-B3 confusion, and s6 errors primarily involved B2-B4. Residual class-error structure therefore differed across targets.



Figure 5. Full-cohort confusion matrix and subject-by-class outcomes. Every cell in the left panel gives the number of trials for a true class (row) and predicted class (column) across all 300 trials. The right panel gives correctly classified trials as correct/6 for each target and true class.

4.4 Internal evidence status and computational replay

Table 5. Final pipeline performance and verification summary

Quantity	Observed value	Interpretation
Mean trial BA	0.8733	Equal-weight mean across s1-s10
Mean trial macro F1	0.8733	Mean of subject-level macro F1
Window BA	0.8730	Duration-weighted propagation of trial decisions
Subject trial-BA range	0.6000-1.0000	Between-subject variation remained
Perfectly classified targets	6/10	All 30 trials correct for each target
Evaluation size	300 trials, 388 windows	Complete B1/B2/B3/B4/B7 panel
Independent computational replay	All checks passed	Selection, locks, predictions, and metrics reproduced
Evidence status	Internal development performance	Not independent external validation

Subject selections, protocol and prediction hashes, every subject BA, and the cohort mean all matched under independent recomputation. Direct access to evaluated-query labels or glove was not detected during the locked run. However, earlier outcomes from the same DB5 cohort had informed the broader research program and candidate-family design; 0.8733 is therefore retrospective internal development performance rather than an independent validation estimate.

5. Discussion

5.1 Principal findings

The final pipeline combined target A1-A12+B5/B6 calibration, other-nine pose semantics, exact-capacity full-batch clustering, and a common target-A stability selector. It achieved mean trial BA and macro F1 of 0.8733 and window BA of 0.8730 on unseen B1/B2/B3/B4/B7 in NinaPro DB5. Subject BA ranged from 0.6000 to 1.0000, and six targets had all evaluated trials classified correctly.

The same 27 candidates and eligibility thresholds were applied to every target. Each target used its own permitted A+B5/B6 data for selection, so final configurations could differ, and the same configuration could recur across targets. The result therefore came from a common adaptation rule that chooses subject-specific representations rather than a manually customized procedure.

5.2 Between-target variability and residual errors

Target-level BA ranged from 0.6000 to 1.0000. Six targets were perfectly classified, while the remaining four did not share one error pair. Cohort means should therefore be accompanied by the target-level performance distribution and class-specific error structure.

The analysis cannot determine physiological or measurement causes for these descriptive differences. Electrode contact, contraction strategy, repetition variability, and movement execution cannot be separated by the present analysis; causal explanation would require dedicated signal-quality and experimental measurements.

5.3 Role of pose semantics

Pose enters the pipeline at three points. The target A+B5/B6 calibration data train the EMG-to-pose bridge; other-nine evaluated-class poses provide semantic class prototypes; and predicted pose can contribute weakly to the clustering representation. Actual target evaluated-query pose is never used for inference.

This organization shares EMBridge's broad premise that pose can structure EMG semantics [5], but the implementation and evidence differ. EMBridge uses learned masked autoencoders, a Q-Former, contrastive alignment, masked reconstruction, and community-aware loss. The present method uses hand-engineered features, linear regression, constrained clustering, and prototypes with explicit intermediate scores. It is a separate structured method, not a reproduction or direct evaluation of EMBridge. No runtime, parameter-count, memory, or formal interpretability comparison was performed.

5.4 Scope of the zero-shot claim

There are zero labeled target EMG examples from B1/B2/B3/B4/B7, which provides an evaluated-class zero-EMG-shot component. The method nevertheless uses other-person pose prototypes for

those classes, a known five-class catalog, the complete 30-trial target batch, and exactly six trials per class. It does not classify trials independently or online.

The most accurate designation is therefore **pose-assisted, closed-catalog transductive recognition of unseen target gestures**. This phrasing acknowledges the meaningful absence of evaluated-class target labels without implying an inductive, open-set, or calibration-free system.

5.5 Interpretation of internal development performance

The locked stability-selection execution did not read evaluated-query labels, and predictions were hash-locked before outcome access. Direct test-label leakage was not detected in that execution. Nonetheless, the candidate family and hypothesis were developed after earlier analyses had exposed outcomes from the same DB5 cohort. Run-level information-access controls cannot erase study-level developmental exposure.

The observed 0.8733 therefore describes reproducible performance of the final pipeline in the current DB5 cohort. It does not independently estimate expected performance for new users or sessions. A candidate family frozen before access to genuinely new outcomes is required for that purpose. This distinction separates procedural reproducibility of the final pipeline from external validity of its performance estimate.

5.6 Limitations

The analysis included only ten intact participants from one public dataset, all of whom informed the development program. Exact-capacity clustering assumes six trials per class. Full-batch transduction does not establish online performance. Although the evaluated target query contributes EMG only, calibration requires target B5/B6 glove and semantic naming requires other-nine evaluated-class glove. The A-to-B split is an operational semantic design, not a universal physiological complexity ordering. The stability family and subject-specific default candidates were retrospectively defined, and a default-candidate rule for a genuinely new target has not been externally validated. Amputee users, different hardware, new sessions, electrode shift, real-time latency, unequal frequencies, unknown class frequencies, and open-set rejection were not tested. Target folds also share participants through overlapping other-nine donor banks and are not fully independent population samples.

5.7 Independent confirmation

A confirmation study should freeze the 27 candidates, stability equation, eligibility thresholds, assignment-margin gate, outcomes, and stopping rules with code hashes before acquiring or opening new outcomes. The locked pipeline should then be evaluated once on participants or sessions whose results were never inspected during method development. Repeating LOSO on the already exposed DB5 s1-s10 cannot restore complete external independence.

Practical follow-up should relax the exact-count prior, evaluate partial and unequal query batches, process sequential trials, reduce the number of pose donors, remove target B5/B6 glove, and quantify latency and session drift. Success criteria should include worst-subject performance, class recall, and error structure in addition to mean BA.

6. Conclusion

In a fully symmetric NinaPro DB5 evaluation, the final pipeline combined target A1-A12+B5/B6 calibration, Target-A stability selection, other-nine pose prototypes, and exact-capacity balanced clustering. It achieved mean trial BA of 0.8733 on unseen B1/B2/B3/B4/B7 while evaluated-query inference used target EMG only. Confusion matrices with trial counts and all subject-level outcomes are reported. Because the same DB5 cohort informed development, this value is internal development performance rather than independent external validation. Online, unknown-frequency, and new-user or new-session performance remain open questions.

Declarations

Ethics approval and consent to participate

This study was a secondary analysis of publicly released, de-identified NinaPro DB5 data and involved no new participant recruitment or signal acquisition. The responsible authors must verify the original-acquisition ethics and consent wording in the dataset publication [3] and determine whether their institution or target venue requires an additional secondary-analysis statement.

Funding

[AUTHOR CONFIRMATION REQUIRED: insert actual funding, grant identifiers, and sponsor roles]

Competing interests

[AUTHOR CONFIRMATION REQUIRED: insert declarations for all authors]

Author contributions

[AUTHOR CONFIRMATION REQUIRED: finalize authorship and CRediT roles]

Data and code availability

The original NinaPro DB5 data remain available under the conditions of the Zenodo dataset record [4]. Analysis code, protocol hashes, and non-identifying derived results will be described at [INSERT REPOSITORY URL/DOI]. The responsible authors must verify the actual release scope and reproduction instructions before submission.

Use of artificial intelligence assistance

OpenAI Codex assisted with local analysis execution, provenance packaging, figure and manuscript drafting, and document formatting. The tool is not an author. Human authors remain accountable for every numerical result, code path, interpretation, citation, authorship decision, and final statement. This disclosure must be adapted to the current policy of the target venue.

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Items Requiring Accountable-Author Completion

1. Finalize author names, affiliations, corresponding author, and CRediT roles from actual contributions.
2. Verify original-acquisition ethics and consent wording and any institutional secondary-analysis requirements.
3. Complete funding, competing-interest, code-release, data-access, and repository statements with actual information.
4. Apply the current target-journal word limit, manuscript structure, figure, reference, and AI-disclosure policies.
5. Describe 0.8733 as internal development performance from the same DB5 cohort, not as an independent external validation estimate.
6. Require human verification of every number, figure, citation, and claim-source pairing before submission approval.